MLE & Sampling Distributions

Quiz review · 10 questions with answers and explanations

Answer Key

Q	Answer
1	2
2	False
3	Increasing the sample size.
4	$\sigma / \sqrt{n}$
5	$(1/n)\ \Sigma\ X_i$
6	It describes how likely the observed data are for given parameter values.
7	$p(1 - p) / n$
8	It must be negative at the candidate point.
9	0.5
10	It gets closer to θ.

Question 1 · Multiple Choice

Suppose the length of fish caught is modeled as normally distributed with mean μ and variance σ^2, where both parameters are unknown. You collect a sample of size n = 10 and calculate the MLEs $\hat{\mu} = 5$ and $\hat{\sigma}^2 = 4$. What is the MLE for the standard deviation of fish length?

- ○ A. $\sqrt{(4/10)}$
- ○ B. 4/10
- ✔ **C. 2**
- ○ D. 4

Answer: 2

By the **invariance property of MLEs**, if θ̂ is the MLE of θ, then g(θ̂) is the MLE of g(θ). Since $\sigma = \sqrt{\sigma^2}$:

$$\hat{\sigma} = \sqrt{\hat{\sigma}^2} = \sqrt{4} = 2$$

The 1/n factor is already built into $\hat{\sigma}^2$, so dividing by 10 again is wrong. $\sqrt{(4/10)}$ looks more like a standard error, which is a different quantity.

Question 2 · True or False

An important property that MLEs have is that they are always unbiased.

- ○ A. True
- ✔ **B. False**

Answer: False

MLEs are not guaranteed to be unbiased. The normal variance MLE divides by n:

$$\hat{\sigma}^2 = (1/n)\,\Sigma\,(X_i - \bar{X})^2,\ E[\hat{\sigma}^2] = ((n-1)/n)\,\sigma^2$$

so it underestimates σ^2 (which is why s^2 uses $n - 1$).

Under regularity conditions, MLEs are **consistent**, **asymptotically unbiased**, **asymptotically normal**, and **asymptotically efficient**. Invariance also breaks unbiasedness in general, since $E[\sqrt{X}] \neq \sqrt{E[X]}$.

Question 3 · Multiple Choice

Which of the following factors reduces the variance of a sampling distribution?

- ✔ **A. Increasing the sample size.**
- ○ B. Increasing the population size.
- ○ C. Decreasing the sample size.
- ○ D. None of the above.

Answer: Increasing the sample size.

$$\mathrm{Var}(\bar{X}) = \sigma^2 / n$$

Larger n shrinks the variance; smaller n inflates it. Population size does not appear in the standard formula. (With the finite population correction, a larger population would slightly *increase* the variance, not reduce it.)

Question 4 · Multiple Choice

What is the standard error of the sample mean for n i.i.d. observations $X_1, \ldots, X_n$ from $N(\mu, \sigma^2)$?

- ○ A. $\sigma^2 / \sqrt{n}$
- ○ B. σ^2 / n
- ○ C. σ / n
- ✔ **D. $\sigma / \sqrt{n}$**

Answer: $\sigma / \sqrt{n}$

The standard error is the standard deviation of the estimator's sampling distribution:

$$SE(\bar{X}) = \sqrt{\mathrm{Var}(\bar{X})} = \sqrt{(\sigma^2/n)} = \sigma / \sqrt{n}$$

Watch out for σ^2/n: that is the *variance* of $\bar{X}$, not its standard error.

Question 5 · Multiple Choice

Suppose a (possibly weighted) die is rolled n times, and the probability of rolling a six is θ. The outcomes are recorded as $X_1, \ldots, X_n$ where $X_i = 1$ if a six is rolled and 0 otherwise. What is the MLE for θ?

- ○ A. $(\Sigma X_i) / (6n)$
- ○ B. None of the above
- ○ C. $(6/n) \Sigma X_i$
- ✔ **D. $(1/n) \Sigma X_i$**

Answer: $(1/n) \Sigma X_i$

Each $X_i \sim$ Bernoulli(θ), so

$$L(\theta) = \theta^{\Sigma X_i} (1 - \theta)^{n - \Sigma X_i}$$

$$\ell'(\theta) = \Sigma X_i/\theta - (n - \Sigma X_i)/(1 - \theta) = 0 \Longrightarrow \theta = \Sigma X_i / n$$

The MLE is just the proportion of sixes. The "6" options are distractors: θ is already defined as P(six), so the number of faces never enters.

Question 6 · Multiple Choice

What is the role of the likelihood function in MLE?

- ○ A. It provides a Bayesian posterior distribution.
- ○ B. It evaluates the probability of parameters given the observed data.
- ✔ **C. It describes how likely the observed data are for given parameter values.**
- ○ D. It minimizes the data's variance.

Answer: It describes how likely the observed data are for given parameter values.

$$L(\theta) = f(x_1, \ldots, x_n \mid \theta), \text{ viewed as a function of } \theta$$

MLE picks the θ that makes the observed data most probable. "Probability of parameters given data" is P(θ | data), the Bayesian posterior, which requires a prior. The likelihood goes the other direction and does not need to integrate to 1 over θ.

Question 7 · Multiple Choice

If $X_1, \ldots, X_n$ are i.i.d. Bernoulli random variables with parameter p, what is the variance of the sampling distribution of the sample proportion $T = (1/n)\ \Sigma\ X_i$?

- ✔ **A. p(1 − p) / n**
- ○ B. p / n
- ○ C. p(1 − p)
- ○ D. p / ((1 − p) n)

Answer: p(1 − p) / n

$\mathrm{Var}(X_i) = p(1 - p)$, and by independence:

$$\mathrm{Var}(T) = (1/n^2) \cdot n \cdot p(1 - p) = p(1 - p) / n$$

Same σ^2/n pattern as Q3, with $\sigma^2 = p(1 - p)$. The p(1 − p) option is the variance of a single observation.

Question 8 · Multiple Choice

Assuming no edge cases, what condition must the second derivative of the log likelihood satisfy to ensure the candidate MLE is a maximum?

- ○ A. It must be positive at the candidate point.
- ○ B. It must be undefined.
- ○ C. It must equal zero.
- ✔ **D. It must be negative at the candidate point.**

Answer: It must be negative at the candidate point.

Second derivative test: $\ell'(\theta) = 0$ finds a critical point, and $\ell''(\theta) < 0$ means the log likelihood is concave down there, so θ is a maximum. Positive means a minimum; zero is inconclusive.

Bernoulli check: $\ell''(\theta) = -\Sigma X_i/\theta^2 - (n - \Sigma X_i)/(1 - \theta)^2 < 0$

Question 9 · Multiple Choice

In estimating the mean height of trees, a forester takes 40 measurements. If the true population distribution is N(20, 25), what is the probability that the sample mean is less than the true mean?

- ○ A. None of the above.
- ○ B. 0.68
- ○ C. 0.95
- ○ D. 0.99
- ✔ **E. 0.5**

Answer: 0.5

$\bar{X} \sim N(20, 25/40)$, $P(\bar{X} < 20) = P(Z < 0) = 0.5$

$\bar{X}$ is centered exactly at μ and the normal is symmetric. The sample size and variance only change the spread, not the center. 0.68 / 0.95 / 0.99 are empirical-rule distractors.

Question 10 · Multiple Choice

The maximum of n i.i.d. random variables $X_1, ..., X_n$ from a uniform distribution over [0, θ] is used to estimate θ. What happens to the expectation of this estimator as n increases?

- ○ A. The behavior is different for different values of θ.
- ✔ **B. It gets closer to θ.**
- ○ C. It gets further from θ.
- ○ D. It is equal to θ.
- ○ E. It stays constant.

Answer: It gets closer to θ.

For $M = \max(X_i)$: $P(M \le m) = (m/\theta)^n$, so $f(m) = n\, m^{n-1}/\theta^n$ on [0, θ].

$E[M] = \int_0^\theta m \cdot n\, m^{n-1}/\theta^n \, dm = (n/(n+1))\, \theta \to \theta$

Biased low for every finite n (the max can never exceed θ), but asymptotically unbiased. This max is the MLE for θ, another example for Q2. Scaling by (n+1)/n makes it unbiased.